

Fairness-Aware Student Dropout Prediction: A Machine Learning Pipeline with Explainability and Bias Mitigation

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Abstract—Dropout risk tools on campus usually report a single accuracy score. They rarely show whether scholarship students are flagged less often than others, or whether male students appear on alert lists twice as often as female students. I analyzed the full Portuguese UCI dataset ($N=4,424$) [1] with four classifiers, a binary early-warning model, Fairlearn group tables, SHAP plots, and two gender mitigators. XGBoost achieved the best macro-F1 (0.711) and accuracy (77.6%). Aid status showed larger gaps than gender: scholarship demographic parity (DP) was 0.264 versus 0.156 for gender, and equal opportunity (EO) was 0.247 versus 0.002 for gender. Second-semester grades dominated SHAP, but tuition and aid fields correlated with sensitive attributes ($|r|$ up to 0.81). Tightening demographic parity for gender reduced accuracy and worsened equal opportunity. Shared dropout benchmarks should report who receives alerts, not only how often the model is correct overall.

Index Terms—student dropout prediction, educational data mining, algorithmic fairness, XGBoost, SHAP, Fairlearn

I. INTRODUCTION

Most early-warning lists are spreadsheets sorted by probability. Advisors rarely see how many scholarship holders are included, or whether male students are flagged at twice the rate of female students. That gap matters. Dropout is costly for students and institutions [1], and models built from enrollment data and first-year grades can flag risk well before withdrawal [2], [3]. Much of the literature still reports cohort-level accuracy [3]–[5]. Fairness studies exist [6]–[8], but they seldom target the dropout datasets researchers actually download and reuse.

Realinho et al. [1] published the Portuguese polytechnic records (UCI ID 697). Martins et al. [2] later showed that boosted trees outperform older methods on the same dataset. Online repositories reproduce those accuracy scores, yet I could not find one study that trains on this split, reports gender and scholarship slices, explains predictions, and tests a mitigator in a single pipeline. Wagner et al. [9] partially addressed this on German enrollment data; Rzepka et al. [10] did so for MOOC logs. I aimed to complete that audit before treating accuracy in the mid-80% range as sufficient.

The work follows a chain of seven notebooks: explore, split, benchmark four models, collapse labels for binary dropout, run Fairlearn, plot SHAP, and test two gender mitigators [11]–[14]. Each notebook writes CSV files that the next notebook

uses. Scholarship gaps exceeded gender gaps on every metric I recorded. No single mitigator improved both DP and EO. The cohort is well-suited to this study because it is public, documented, and widely cited in published accuracy leaderboards, yet slices by aid status are rarely reported alongside headline F1 scores.

II. RELATED WORK

Delen [3] compared trees and ensembles on university logs but reported only overall accuracy. Márquez-Vera et al. [5] applied genetic programming to imbalanced school data in the same way. Shahiri et al. [4] reviewed the field later and pointed out how seldom anyone breaks down errors by student type. Hu and Rangwala [8] extended this line of work: at-risk models can pick up historical bias, and they tested ways to limit sensitivity to protected inputs.

On UCI ID 697, Realinho et al. [1] describe the schema; Martins et al. [2] add SMOTE and strong XGBoost results but no group-level tables. Gardner et al. [6] argue for slice-based reviews in learning analytics. Baker and Hawn [7] trace how socioeconomic status enters through ordinary academic fields. Wagner et al. [9] fit logistic models on German data with uneven slices by program and sex, which is the closest prior work to my gender- and scholarship-based analysis of the Portuguese dataset.

This pipeline uses XGBoost [15], SHAP [14], and Fairlearn [13] with demographic parity and equalized odds [11], [12]. I did not propose a new learner; the contribution is to connect established tools on a fixed public split so others can reproduce the fairness tables. Table I compares that setup with prior dropout and fairness studies.

III. DATASET AND METHODS

A. Dataset and Preprocessing

All experiments use UCI ID 697 [1], [2]. Table II lists 4,424 students, 36 inputs, three outcomes (dropout 32.1%, enrolled 17.9%, graduate 49.9%), and no missing values. Records cover 2008/2009 through 2018/2019 across 17 degree programs. The cohort is 64.8% female, 24.8% of students hold scholarships, and 97.5% are Portuguese nationals. Age at entry ranges from 17 to 70 years, with a median of 20.

TABLE I: How this study extends prior dropout and fairness work (✓ = addressed; — = not reported in source). Wagner et al. list stakeholder explanations as future work. Rzepka et al. report fairness audits without post-hoc mitigation.

Study	Same UCI data	Subgroup audit	SHAP + proxy check	Mitigation test
Martins et al. [2]	✓	—	—	—
Realinho et al. [1]	✓	—	—	—
Gardner et al. [6]	—	✓	—	—
Wagner et al. [9]	—	✓	—	—
Rzepka et al. [10]	—	✓	—	—
This work	✓	✓	✓	✓

TABLE II: Dataset summary (full cohort, $N=4,424$).

Statistic	Value
Input features	36
Dropout / Enrolled / Graduate	32.1% / 17.9% / 49.9%
Female students	64.8%
Scholarship holders	24.8%
Portuguese nationality	97.5%
Age at enrollment (median)	20
Missing values	0

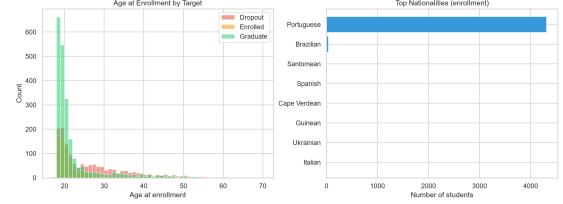


Fig. 3: Age at enrollment distribution and nationality composition (EDA on full cohort).

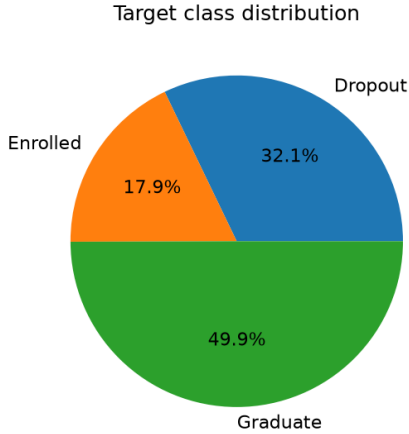


Fig. 1: Target class distribution across the full cohort.

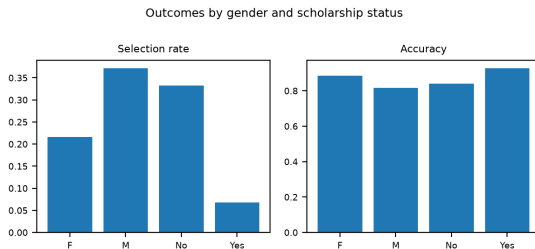


Fig. 2: Outcome mix by gender and scholarship on the full cohort (EDA).

The enrolled class is the smallest share of the three labels (Fig. 1). That skew is why macro-F1 penalizes models that ignore the middle class.

I plotted outcome mix by gender and scholarship on the full cohort before model training (Fig. 2). Graduates dominate every bar, but the dropout share shifts enough between groups to justify separate alert-rate checks later.

Age and nationality plots (Fig. 3) show almost no nationality

variation; I kept both fields for the fairness audit nonetheless.

Categorical columns were one-hot encoded, and numeric columns were standardized. A stratified 80/20 split with seed 42 produced 3,539 training rows and 885 test rows; all fairness and mitigation results come from the test fold. Gender, Nationality¹, Scholarship holder, and Age at enrollment were stored in separate files for auditing and were excluded from the feature matrix.

B. Models, Fairness Framing, and Mitigation

Four models were compared on the training split with stratified five-fold CV: multinomial logistic regression (L-BFGS, 1000 iterations), random forest (300 trees), XGBoost (300 trees, depth 6, learning rate 0.1) [15], and a 128→64 MLP with ReLU, Adam, and early stopping (10% holdout, patience 10). I left out SMOTE and class weights to stay close to Martins et al. [2]. Reported scores are accuracy, macro-F1, and macro OvR AUC.

Fairlearn expects a binary outcome [12], so *Enrolled* and *Graduate* were merged into one non-dropout label, following the early-warning framing used by Wagner et al. [9]. The best XGBoost setup was refit with `objective=logistic` on those binary labels. `MetricFrame` then produced per-group accuracy, F1, selection rate, and DP/EO for gender and scholarship.

TreeSHAP [14] on 500 stratified test rows ranked mean |SHAP|; I also computed Pearson r between top features and sensitive fields. Mitigation tests focused on gender: Exponentiated Gradient with DP $\epsilon=0.02$ [11], plus ThresholdOptimizer under equalized odds on a pre-fitted logistic model [12], [13]. Scholarship mitigation is reserved for a separate run; the gaps on aid status were larger and need their own tuning budget.

Code ran in Python 3.12 with scikit-learn, XGBoost 3.3, Fairlearn 0.14, and SHAP 0.52. CV used `StratifiedKfold`

¹Original UCI field name.

($k=5$, seed 42). Every figure in this paper traces to a saved CSV.

C. Fairness Metrics

Let $Y \in \{0,1\}$ denote the true dropout label and $\hat{Y}$ the model prediction. For sensitive attribute A and group value a , the *selection rate* is $P(\hat{Y}=1 \mid A=a)$. *Demographic parity difference* is

$$DP = \max_a P(\hat{Y}=1 \mid A=a) - \min_a P(\hat{Y}=1 \mid A=a). \quad (1)$$

Equal opportunity difference compares true-positive rates among students who actually dropped out:

$$EO = \max_a P(\hat{Y}=1 \mid Y=1, A=a) - \min_a P(\hat{Y}=1 \mid Y=1, A=a). \quad (2)$$

Definitions follow Fairlearn [12], [13], consistent with recent educational fairness studies [9], [10]. Lower values are better for both metrics. In these runs, the two metrics often moved in opposite directions after mitigation, which is why both remain in the tables.

Only 205 scholarship holders appear in the test fold, and the enrolled class is thin overall. The group numbers are still worth reporting, but they should be read with that imbalance in mind.

IV. RESULTS

A. Predictive Performance

Table III and Fig. 4 summarize a five-fold CV on the training data. XGBoost achieved the best macro-F1 (0.711) and accuracy (77.6%). Logistic regression and the MLP trailed by a few points. Random forest achieved the highest AUC (0.886) but the lowest macro-F1 (0.658), largely because it often mislabeled enrolled students on this imbalanced label mix.

TABLE III: Baseline comparison (5-fold stratified CV, training set).

Model	Acc.	F1 (macro)	AUC
XGBoost	0.776	0.711	0.885
Logistic Regr.	0.766	0.698	0.881
MLP (2-layer)	0.768	0.691	0.882
Random Forest	0.765	0.658	0.886

TABLE IV: Cross-validation mean $\pm$ std (training set, 5 folds).

Model	Acc.	F1 (macro)	AUC
XGBoost	0.776 ± 0.012	0.711 ± 0.013	0.885 ± 0.011
Logistic Regr.	0.766 ± 0.010	0.698 ± 0.011	0.881 ± 0.010
MLP (2-layer)	0.768 ± 0.011	0.691 ± 0.026	0.882 ± 0.011
Random Forest	0.765 ± 0.012	0.658 ± 0.022	0.886 ± 0.010

Fold variance stayed tight (Table IV): XGBoost macro-F1 ranged 0.700–0.734 (± 0.013). Table V details each XGBoost fold; Table VI does the same for every baseline. The MLP was noisier (± 0.026 on F1) and dropped to 0.650 on fold 2, so I did not carry it into the fairness stage. On the held-out binary task, XGBoost reached 86.0% accuracy and an F1 of 0.763, in line with prior work on this dataset [1]. The remainder of this section focuses on slices rather than the leaderboard.

TABLE V: XGBoost per-fold CV scores (training set).

Metric	F1	F2	F3	F4	F5
Accuracy	0.775	0.774	0.798	0.768	0.764
F1 (macro)	0.715	0.700	0.734	0.705	0.701
AUC	0.880	0.889	0.905	0.875	0.878

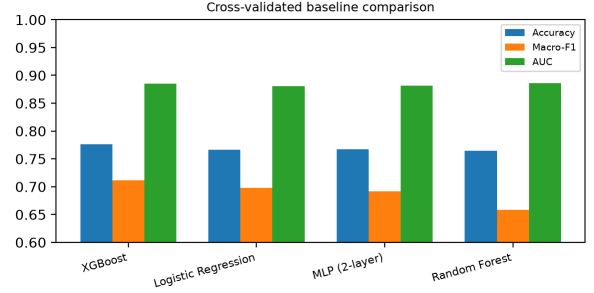


Fig. 4: Cross-validated accuracy, macro-F1, and AUC across baseline models.

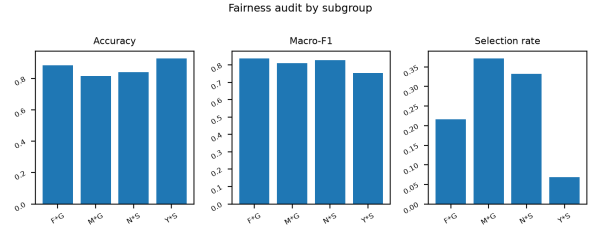


Fig. 5: Accuracy, macro-F1, and selection rate by gender and scholarship group.

B. Fairness Audit

Table VII splits binary XGBoost predictions by group on the test fold. Males were flagged 37.1% of the time versus 21.6% for females. Scholarship showed a sharper contrast: holders reached 92.7% accuracy, but only 6.8% were flagged; non-holders reached 84.0% accuracy with 33.2% flagged.

Table VIII summarizes those as dataset-level gaps: $DP=0.156$ and $EO=0.002$ for gender; $DP=0.264$ and $EO=0.247$ for scholarship. The scholarship panel in Fig. 5 is the one I would put in front of a registrar: high accuracy among students receiving aid, paired with almost no alerts.

Across 885 test rows, the model flagged 123 females and 117 males—similar counts, but very different rates because the fold has 570 women and 315 men (Table IX). Only 14 of 205 scholarship holders were flagged, against 226 of 680 non-holders. Table X lists every gap Fairlearn exported; scholarship gaps exceed gender gaps on all five columns, including an F1 gap of 0.074.

Fig. 5 visualizes the same group-level accuracy, macro-F1, and selection-rate patterns. Scholarship holders combine high accuracy with the lowest alert rate on the test fold.

C. Explainability

The SHAP summary (Fig. 6) ranks second-semester curricular grade first ($|SHAP|=1.30$), then tuition-fee status, age at enrollment, and admission grade. *Course_171* (Animation and Multimedia Design in the UCI notes) also ranked high.

TABLE VI: Per-fold cross-validation macro-F1 (training set, stratified 5-fold).

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean $\pm$ std
XGBoost	0.715	0.700	0.734	0.705	0.701	0.711 $\pm$ 0.013
Logistic Regr.	0.711	0.689	0.712	0.686	0.690	0.698 $\pm$ 0.011
MLP (2-layer)	0.716	0.650	0.722	0.680	0.689	0.691 $\pm$ 0.026
Random Forest	0.689	0.632	0.669	0.635	0.666	0.658 $\pm$ 0.022

TABLE VII: Fairness by group (binary dropout task, test set).

Attribute	Group	Acc.	F1	Sel. Rate	n
Gender	Female	0.884	0.837	0.216	570
	Male	0.816	0.811	0.371	315
Scholarship	No	0.840	0.826	0.332	680
	Yes	0.927	0.753	0.068	205

TABLE VIII: Disparity summary (test set).

Sensitive attribute	DP diff.	EO diff.
Gender	0.156	0.002
Scholarship holder	0.264	0.247

TABLE IX: Binary dropout alerts on the held-out test set ($n=885$).

Attribute	Group	n	Flagged	Rate
Gender	Female	570	123	21.6%
	Male	315	117	37.1%
Scholarship	No	680	226	33.2%
	Yes	205	14	6.8%

TABLE X: Full fairness disparity metrics (binary task, test set).

Attribute	Acc. gap	F1 gap	Sel. gap	DP diff.	EO diff.
Gender	0.068	0.026	0.156	0.156	0.002
Scholarship holder	0.087	0.074	0.264	0.264	0.247

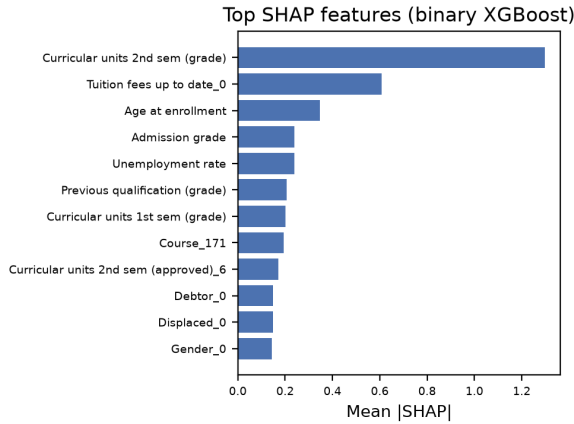


Fig. 6: Top SHAP features for the binary XGBoost dropout model (500 test samples).

Table XI lists the top ten; second-semester grade alone outweighs the next three combined.

Scholarship- and gender-related dummies appeared among the top 15 features. Table XII keeps Pearson pairs above the 0.30 cutoff from the notebooks. `Scholarship holder_0` against actual aid status yielded $r=0.81$; the correlation

TABLE XI: Top ten SHAP features by mean |SHAP| (500 test samples).

Feature	SHAP
Curricular units 2nd sem (grade)	1.298
Tuition fees up to date_0	0.609
Age at enrollment	0.347
Admission grade	0.240
Unemployment rate	0.238
Previous qualification (grade)	0.207
Curricular units 1st sem (grade)	0.202
Course_171	0.194
Curricular units 2nd sem (approved)_6	0.172
Debtor_0	0.150

between age SHAP and age at enrollment was 0.67. Martins et al. [2] report the same grade dominance on this dataset; Baker and Hawn [7] would treat the tuition and aid links as proxy paths worth monitoring.

TABLE XII: Strongest SHAP-sensitive-attribute correlations (proxy screen).

Feature	Sensitive field	$ r $
Scholarship holder_0	Scholarship status	0.81
Age at enrollment	Age	0.67
Gender_0	Gender	0.30

D. Mitigation Trade-offs

Gender mitigation results appear in Table XIII and Fig. 7. Exponentiated Gradient reduced DP from 0.156 to 0.016 but lowered accuracy by 2.9 percentage points and increased EO to 0.274. ThresholdOptimizer had a smaller impact: DP 0.087, EO 0.071, and accuracy fell by 0.6 percentage points. On a live campus, I would try threshold tuning before heavier methods.

TABLE XIII: Gender mitigation on the test set (before/after and Δ vs. baseline XGBoost).

Model	Acc.	F1	DP	EO	Δ Acc.	Δ F1	Δ DP	Δ EO
Before (XGBoost)	0.860	0.763	0.156	0.002	—	—	—	—
After (Exponentiated Gradient)	0.831	0.707	0.016	0.274	-0.029	-0.056	-0.140	+0.272
After (Threshold Optimizer)	0.854	0.752	0.087	0.071	-0.006	-0.011	-0.069	+0.070

Exponentiated Gradient reduced DP by 0.140 ($\sim 90\%$ relative) but lowered accuracy by 0.029 and increased EO by 0.272. ThresholdOptimizer was less disruptive—for a first deployment, I would start with threshold adjustment; Exponentiated Gradient only makes sense if matching flag rates across gender is a hard requirement.

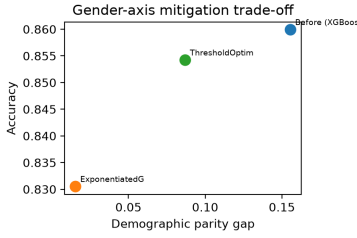


Fig. 7: Accuracy versus demographic parity gap under gender-axis mitigation.

V. DISCUSSION

Gardner et al. [6] and Wagner et al. [9] both warn that headline accuracy can hide weak slices. I found the same pattern in the Portuguese cohort. Scholarship students achieved the best accuracy and the fewest flags, a combination that could hide struggling aid recipients if staff only read the overall score.

TABLE XIV: Headline numbers for quick audit review (test fold unless noted).

Outcome	Value
Best CV macro-F1 (XGBoost, train)	0.711
Binary test accuracy (XGBoost)	0.860
Binary test F1 (XGBoost)	0.763
Gender DP difference	0.156
Scholarship DP difference	0.264
Scholarship alerts ($n=205$)	14

On gender, the picture differed. EO stayed near zero, so the tension was *who* got flagged (DP 0.156), not *who* among true dropouts was missed. Reporting one fairness number would hide that. Forcing DP down with Exponentiated Gradient sent EO to 0.274, as Rzepka et al. [10] report in other settings. Campus administrators must decide which error they can tolerate before picking a mitigator.

SHAP clarified part of the mechanism. Grades and tuition are plausible inputs, but when scholarship SHAP values correlate with true aid status at 0.81, the model is partly reading socioeconomic signal through ordinary fields [7]. Removing the scholarship column from the feature list would not necessarily break that link.

Dropout papers on public files should, at minimum, publish slice tables beside accuracy, state clearly whether the label is binary or three-way, and show mitigation trade-offs instead of calling one model “fair.” Shahiri et al. [4] reviewed many algorithm comparisons with no fairness tables at all. I kept four baselines so the audit does not depend on a single tuned tree. Fixed splits on UCI ID 697 [1] matter too—without them, DP figures from different papers do not align [4]. This work sits next to Martins et al. [2] on accuracy, and next to Gardner et al. [6] and Wagner et al. [9] on subgroup reporting. The headline audit numbers in Table XIV are intended as a compact checklist for that comparison.

A. Implications for Advising Practice

For a campus team reading these tables, a practical sequence helps: check the slice table before trusting the global score; confirm whether the intervention is truly binary dropout or

three-way status; pick DP or EO (or both) in writing before any threshold move; and keep an appeals path if scores ever reach students [7], [8]. Nothing here is a deployment preset—Section IV is audit material. No probability replaces advisor judgment. Early-warning scores should support outreach, not automatic sanctions.

Scholarship students illustrate the point: 92.7% accuracy with only 14 alerts is exactly when a separate review path makes sense. A practical dual-path workflow is to keep the model score for general cohort triage while flagging aid recipients with falling second-semester grades for manual review, even when the model probability stays low. That step does not require retraining; it responds directly to the largest parity gap in Table XIV. Hu and Rangwala [8] argue for subgroup monitoring before any live rollout; the Portuguese audit shows why aid-status slices deserve that treatment before gender-only mitigation is discussed.

VI. CONCLUSION

The scope has clear limits. The data come from one Portuguese polytechnic with a predominantly domestic student body, so retention models may not transfer cleanly to other campuses [3], [5]. The analysis focused on gender and scholarship; nationality and age were not examined in depth. Merging enrolled and graduate labels matches common advising practice but removes outcome detail. Mitigation stopped at gender, even though scholarship gaps were worse. There was no nested hyperparameter search, no year-by-year holdout, and SHAP correlations are descriptive only.

XGBoost still reached 86.0% accuracy and an F1 of 0.763 while flagging only 14 scholarship students out of 205. Strong averages can mask bad slices. Fairlearn and SHAP make that visible and show what accuracy you give up when DP is tightened.

Everything traces to public UCI data, seed 42, and exported CSVs from seven notebooks. Another team can rerun the chain and verify the results. For a camera-ready version, I would publish the exact train and test row indices so fairness tables stay comparable without re-deriving splits. Tables V, VI, and IX are the three I would share with a review board before any model is deployed.

Scholarship-axis mitigation is next on the list (EO is already 0.247), along with multi-attribute constraints, ablations on tuition and aid proxies, and a second institution to see whether the aid pattern repeats. The main takeaway from this run is simpler: a model can appear strong on accuracy metrics while still failing the students who need outreach most.

For teams reusing UCI ID 697, I recommend publishing four artifacts with any new accuracy claim: the stratified split indices, group-level selection rates, a SHAP or proxy screen for aid and gender fields, and at least one mitigation trade-off table. That minimum package would make fairness comparisons across papers possible without repeating the full notebook chain.

ACKNOWLEDGMENTS

The author thanks Prof. Dr. Jarin T, Jyothi Engineering College, Thrissur, Kerala, India, for academic guidance and review of

this work.

DATA AND CODE AVAILABILITY

The UCI Portuguese dataset is open (repository ID 697) [1]. Train and test partitions use seed 42 with stratified sampling. The reproducible pipeline comprises seven notebooks: (1) exploratory data analysis, (2) stratified train–test split, (3) baseline model benchmarking, (4) Fairlearn fairness audit, (5) SHAP explainability and proxy screening, (6) gender-axis mitigation, and (7) compilation of tables and figures. Each stage writes CSV outputs under `results/` that the next stage reads, so every metric in this paper traces to a saved file. Notebook outputs are versioned together with the figures in `paper/figures/`.

LLM USAGE STATEMENT

Language models helped with early notebook layout and draft wording. Every number, table, figure, and reference was checked manually against executed code and saved outputs. The author accepts responsibility for the final text.

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